THE NOVA RESEARCH CURRICULUM

General Science and Social Studies

Grade 5 · Term 2

Student edition

General Science: SC-5.4 pollution, SC-5.5 physical and chemical changes, SC-5.6 light and sound. Social Studies: SS-5.3 latitude, longitude and grid references; SS-5.4 Pakistan's physical regions, climate, global warming and population.

General Science · lesson notes

- **SC-5.4 (Weeks 1–4):** air pollution (smoke, smog in winter), water pollution (waste in rivers), land pollution (plastic); effects on health and wildlife; reduce, reuse, recycle; a neighbourhood pollution survey.
- **SC-5.5 (Weeks 5–8):** physical changes can usually be reversed and make no new substance (melting ice, cutting paper); chemical changes make a new substance (burning, cooking an egg, rusting). Investigation: what makes iron nails rust?
- **SC-5.6 (Weeks 9–12):** light travels in straight lines (three cards with holes in a line); shadows; reflection with mirrors. Sound is made by vibrations and travels through solids, liquids and gases; build a string telephone.

Social Studies · lesson notes

- **SS-5.3 (Weeks 1–4):** lines of latitude (equator) and longitude (prime meridian); find places by grid reference on a map of Pakistan.
- **SS-5.4 (Weeks 5–12):** physical regions (northern mountains, western plateau and Balochistan, Indus plains, deserts, coastal areas); factors affecting climate (latitude, altitude, distance from sea); global warming and Pakistan's glaciers and floods; population growth and its effects.

Workbook pages

Page	Outcome	Activity
1	SC-5.4	Pollution survey record.
2	SC-5.5	Physical or chemical? Sort 12 changes; rusting fair test.
3	SC-5.6	Straight-line light test; string telephone report.
4	SS-5.3	Grid-reference treasure hunt.
5	SS-5.4	Colour Pakistan's physical regions; a climate-factors table.

Question bank

#	Outcome	Question	Your answer
1	SC-5.4	Smog is a kind of ... pollution.	water / land / air / noise

#	Outcome	Question	Your answer
2	SC-5.4	Plastic bags can harm animals. True or false.	True / False
3	SC-5.4	Give two ways to reduce pollution.	_____
4	SC-5.5	Which is a chemical change?	melting ice / cutting paper / burning wood / folding cloth
5	SC-5.5	Melting butter is a physical change. True or false.	True / False
6	SC-5.5	Iron rusts when it meets ...	sand / water and air / paper / sunlight only
7	SC-5.6	Light travels in ... lines.	curved / straight / zig-zag / circular
8	SC-5.6	Sound is made by ...	light / vibrations / colours / shadows
9	SC-5.6	Sound can travel through a string. True or false.	True / False
10	SS-5.3	The 0° line of latitude is the ...	prime meridian / equator / tropic / border
11	SS-5.3	Longitude lines run from pole to pole. True or false.	True / False
12	SS-5.4	K2 is in Pakistan's ...	Indus plains / northern mountains / Thar desert / coast
13	SS-5.4	Higher places are usually ...	hotter / colder / wetter always / the same
14	SS-5.4	Name one effect of global warming on Pakistan.	_____